NATHANIEL EDWARD-INATIMI

Space Weather PhD Researcher, Astrophysics Masters Graduate

orcid.org github.com

PROFESSIONAL SUMMARY

Award-winning, final-year PhD researcher with an interdisciplinary research profile in space weather, meteorological methods, and statistical uncertainty. Developing uncertainty-aware research tools for solar physics community.

RELEVANT EXPERIENCE

PhD Researcher

Dept. of Meteorology, University of Reading (NERC Scenario DTP)

Sep 2023 – Present *[Est. Feb 2027]* Reading, UK

Thesis: Smarter ensembles for solar wind forecasts.
Supervisors: Prof. Mathew Owens and Dr. Luke Barnard
☑ Calibrating and improving coupled coronal-heliospheric solar-wind ensembles; adapting tried-and-tested meteorological principles into the field of space weather. ❯Received pump-prime funding *(£1500)* to explore hybrid-AI methods for space weather forecasting.
★ Awarded 2025 *Quo Vadis* Seminar Prize for best PhD presentation.
★ Awarded 2025 3rd-year Poster prize for best PhD poster.
Three accepted first-author publications; contributed to peer-review.

Visiting Scientist

UK Met Office

May 2025 Exeter, UK

☑ Explored new ensemble forecasting methods using operational space weather models; migrated operational model (Wang-Sheeley-Arge Fortran model source code) onto recently upgraded HPC systems.

Student Researcher

SENSE CDT Research Experience Placement

June 2022 – August 2022 Edinburgh, UK

☑ Used machine learning to predict the timing, magnitude, and impact of solar flares; processed SDO magnetogram (remote-sensing) data for a neural network. *(Supervisors: Prof. Kathy Whaler and Dr. Ciarán Beggan)*

ADDITIONAL EXPERIENCE

Research Officer

University of Reading Race Equity Group

Nov 2024 – Present Reading, UK

Researcher/Advocate for inclusive research culture (recent work in-prep https://dx.doi.org/10.2139/ssrn.6691900)

Awards Taskforce

Magnetosphere Ionosphere and Solar-Terrestrial

Jan 2026 – Present UK

Group promoting equal representation for award nominations across UK MIST community (see https://mist.ac.uk/awards).

Observing Member for Board of Trustees

V&A Dundee – Scotland's design museum

September 2021 – March 2024 Dundee, UK

Advocate for progressive governance; key representative for the young trustees movement.

PROFESSIONAL DEVELOPMENT

ESA Solar Orbiter (SolO) Hackathon

Nov 2025 – Present Noordwijk, Netherlands

Collaboration on solar connectivity science; now leading development of community tool.

ESA Acceleration Days Hackathon on Space Weather Resilience

June 2025 Darmstadt, Germany

★ Produced winning pitch on interpretable space weather alert systems.

Conference Local Organising Committee

Jan – Sept 2024 Reading, UK

Vice-chair and session convener for 2024 Joint DTP Conference *"Micro to Macro: Changing environments on every scale"*

Scientific Software Development Training

Feb – March 2024 Reading, UK

Scenario DTP training course on core developer skills needed for scientific Python.

STFC Summer School: Introductory Course in Solar and Solar-Terrestrial Physics

August 2023 St Andrews, UK

EDUCATION

MPhys Astrophysics (First Class)

University of Edinburgh

September 2018 – May 2023

★ Awarded 2021 School of Physics Career Development Scholarship *(£1500)*.
★ 2nd year Certificate of Merit for outstanding marks (2019/2020).
★ Edinburgh award recognising volunteer work as peer-mentor (2019/20, 2020/21).

Thesis/Dissertation

Mphys: *"A Study of the Solar Chromosphere"*
BSc: *"Save the Earth from Killer Asteroids!"*

High School

Morgan Academy – Dundee, Scotland

August 2011 – June 2017

KEY SKILLS

Python [8 yrs] | MATLAB [2 yrs] | Fortran [2 yrs] | Statistical uncertainty | Machine Learning | Modelling

PUBLICATIONS

First-author Publications

Edward-Inatimi, N. O., M. J. Owens, L. Barnard, H. Turner, M. Marsh, S. Gonzi, M. Lang, and P. Riley. "Adapting Ensemble-Calibration Techniques to Probabilistic Solar-Wind Forecasting". In: *Space Weather* 22.12 (2024). DOI: https://doi.org/10.1029/2024SW004164

Edward-Inatimi, N. O., M. J. Owens, L. Barnard, H. Turner, M. Marsh, S. Gonzi, and M. Lang. "Calibrating Probabilistic Solar-Wind Forecasts Driven by the Wang-Sheeley-Arge Model". In: *Space Weather* 24.3 (2026). DOI: https://doi.org/10.1029/2025SW004614

Edward-Inatimi, N. O., M. J. Owens, L. Barnard, M. Lang, H. Turner, M. Marsh, S. Gonzi, and A. Yeates. "Adapted Empirical Modeling of Near-Sun Solar-Wind Speeds for Use in Ensemble Forecasts". In: *Space Weather* 24.9 (2026). DOI: https://doi.org/10.1029/2026SW005224

Collaborative Publications

M. Billcliff, A. W. Smith, M. J. Owens, W. L. Woo, L. Barnard, **Edward-Inatimi, N. O.**, and I. J. Rae. "Extended Lead-Time Geomagnetic Storm Forecasting With Solar Wind Ensembles and Machine Learning". In: *Space Weather* 24.3 (2026). DOI: https://doi.org/10.1029/2025SW004823

M. J. Owens, L. A. Barnard, H. Turner, D. Gyeltshen, **Edward-Inatimi, N. O.**, J. O'Donoghue, M. Lockwood, S. Watson, M. Rutala, C. M. Jackman, and P. Riley. "Driving Dynamical Inner-Heliosphere Models With In Situ Solar Wind Observations". In: *Space Weather* 24.1 (2026). DOI: https://doi.org/10.1029/2025SW004675

NOTABLE PRESENTATIONS

Invited talk for the Solar Orbiter Working Group
Oral Presentation
Sep 2026 — Online

Invited talk for UK Met Office
Oral Presentation
June 2026 — Exeter, UK
Collaboration meeting with UKMO Space Weather Research team and NASA Goddard Space Flight Center.

ESA Heliospheric Modelling Workshop
Oral presentation
April 2026 — Darmstadt, Germany
Presentation to key ESA representatives and academics within heliospheric modelling community.

Invited talk for University of Warwick, Centre for Fusion Space and Astrophysics
Oral presentation
Dec 2025 — Warwick, UK

Royal Meteorological Society Public Talk: PhD Showcase
Oral presentation
Oct 2025 — Reading, UK
Public outreach talk presenting PhD work to the general public.
Available to watch.

PERSONAL REFERENCES AVAILABLE ON REQUEST.

CONFERENCE CONTRIBUTIONS

National Astronomy Meeting 2026
Oral presentation
July 2026 — Birmingham, UK

Solar Wind 17
Poster presentation
June 2026 — Bath, UK

European Geophysical Union 2026
Oral Presentation
May 2026 — Vienna, Austria
Abstract link: https://doi.org/10.5194/egusphere-egu26-12367

European Space Weather Week 2025
Oral and Poster presentation
Oct 2025 — Umeå, Sweden

RAL Space Early Careers Conference 2025
Poster presentation
Oct 2025 — Harwell, UK

International Research Culture Conference 2025
Poster presentation
Sep 2025 — Warwick, UK

Spring Magnetosphere, Ionosphere and Solar-Terrestrial Conference 2025
Poster Presentation
April 2025 — Leicester, UK

European Space Weather Week 2024
Oral Presentation
Nov 2024 — Coimbra, Portugal

European Space Weather Week 2022
Virtual Poster Presentation
Oct 2022 — Zagreb, Croatia